Business Recommendations

Customer churn - telecommunications. Companion to the Final Report (method and model details).

1. The situation in numbers

- 1,869 of 7,043 customers (**26.5%**) churned in this sample, carrying **$139,131 per month** of recurring revenue (30.5% of the monthly total).
- Churn is concentrated, not spread evenly: **month-to-month** customers are 55% of the base but **88.6% of churners**; customers in their **first 12 months** churn at 47.4%; fiber-optic customers at 41.9%.
- The low-risk reference group (1- or 2-year contract with automatic payment, 1,934 customers) churns at only **5.7%**.

2. Where to act first (segments overlap - do not add the rows)

Segment	Customers	% of base	Churn rate	% of all churners	Monthly $ lost
Month-to-month contract	3,875	55.0%	42.7%	88.6%	$120,847
Month-to-month, first 12 months	1,994	28.3%	51.4%	54.8%	$68,301
Month-to-month + fiber	2,128	30.2%	54.6%	62.2%	$100,482
Electronic-check payers	2,365	33.6%	45.3%	57.3%	$84,289
Internet, no security and no tech support	2,553	36.2%	49.0%	66.9%	$95,153
Month-to-month + e-check + fiber	1,307	18.6%	60.4%	42.2%	$68,282
1/2-year contract + auto-pay (reference)	1,934	27.5%	5.7%	5.9%	$9,551

'Monthly $ lost' = sum of monthly charges of customers in the segment who churned. These are descriptions of the past, not forecasts.

3. Recommended actions

#	Action	Why (evidence)	How to test it
1	Onboarding programme for new month-to-month customers (check-ins in months 1-6, early-life offers)	51.4% churn in months 0-12 for month-to-month; 1,024 churners (54.8% of all)	Randomly hold back 20% of new customers as a control group; compare 90-day churn
2	Offer a benefit for moving to a 1- or 2-year contract	Churn 42.7% / 11.3% / 2.8% by contract	Same offer to a random half of eligible customers; loyal customers may self-select, so only a control group shows the real effect
3	Diagnose fiber before discounting: service quality, price versus competitors, exit survey	Fiber 41.9% vs DSL 19.0%, and the gap persists within each contract type; month-to-month fiber loses $100,482/month	Exit survey and fault/complaint analysis first; pilot price-lock only after the cause is known
4	Trial free security / tech-support for at-risk internet customers; skip streaming promotions	Without security 41.8% vs 14.6% with; 2,553 customers have neither (49.0% churn); streaming gap only 3-4 points	Randomised 3-6 month free trial; measure churn and later paid take-up
5	Move electronic-check payers to automatic payment with a small incentive	45.3% vs 15.2-16.7% for automatic methods. The model finds little extra signal in payment method, so expect a smaller effect	A/B test the incentive; track conversion to auto-pay and churn
6	Review pricing for high-bill, no-commitment customers	Churn 37.8% in the $70-90 band vs 9.8% at $30 or less; price cannot be separated from service type here	Test a price-lock-for-contract offer on a random subset
7	Run a model-guided tiered campaign (section 4)	Model ranks customers well (test ROC-AUC 0.84)	Keep a random hold-out inside every tier
8	Tailored service for senior customers (not worse terms)	1,142 seniors, 41.7% churn vs 23.6%	Fairness review first; test service changes, not eligibility rules

Not supported by the data: gender-based campaigns (26.9% vs 26.2%); phone-service or streaming promotions as retention levers (small gaps).

4. Using the churn model in the retention team

The model gives every customer a risk score. On the 1,409 held-out customers, the three tiers behaved as follows:

Tier	Rule	Held-out customers	Actually churned
High	score 0.75 or above	288 (20.4%)	66.0%
Medium	0.35 to 0.75	470 (33.4%)	31.3%
Low	below 0.35	651 (46.2%)	5.7%

Suggested use: High tier gets a personal call or a tailored offer; Medium tier gets low-cost automated outreach (email/SMS); Low tier gets normal service. The riskiest 20% hold about half of all churners. Scores are for *ranking*, not calibrated probabilities: a score of 0.6 does not mean a 60% chance.

Is an offer worth it? The cut-offs should come from economics, which this dataset cannot provide. An offer pays off when: (chance the offer actually keeps the customer) x (customer's monthly margin) x (extra months they stay) is greater than the offer cost. Finance should supply margin and offer cost; the pilots in section 3 supply the 'chance the offer works'. The model predicts who is *likely to leave*, not who will *respond* to an offer.

5. Roadmap

- **Weeks 1-2:** agree cost assumptions with Finance and Marketing; set up the control-group process; run exit surveys on fiber leavers.
- **Weeks 3-8:** pilot actions 1, 2 and 4 on randomised groups; score the live customer base with the model each month.

- **Month 3:** compare churn between treated and control groups, calculate cost per retained customer, scale what works and drop what does not.
- **Ongoing:** re-test the model on recent data every quarter and review fairness (age, gender) before wider use.

6. Limits of this analysis

- Associations in a historical sample: nothing here proves that a contract, price or service change *causes* customers to stay.
- No dates or reasons for leaving, so some of the 'short tenure' signal comes from how the data was recorded, and real-world performance may be lower than reported.
- No claim is made about savings or churn reduction; that requires the pilot results above.
- **Data that would sharpen this:** churn reasons, usage and network quality, complaints and support calls, competitor prices, prior offers and responses, customer margin and acquisition cost.